ISC Class XII Physics

Chapter 14
Semiconductor Electronics

3 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

I. p-n Junction Diode – Forward Bias, Barrier Potential & Dynamic Resistance

Compiled Answer Key – Step-by-step ISC board-style solutions

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Formula Sheet

I. p-n Junction Diode – Forward Bias

Formula: $V_{\text{battery}} = I_{\text{max}}R + V_D$
V_{battery} = battery EMF, I_{max} = maximum safe forward current, R = series resistance, V_D = constant forward voltage drop across the diode.

Formula: $R_{\text{dynamic}} = \dfrac{\Delta V}{\Delta I}$
R_{dynamic} = dynamic (AC) resistance of the diode, ΔV = change in voltage across diode, ΔI = corresponding change in diode current.

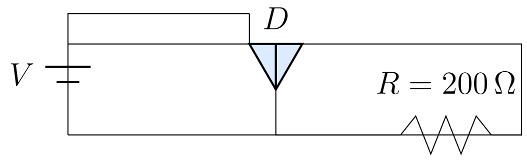
Formula: $I = \dfrac{V - V_0}{R}$
I = current in the circuit, V = applied battery voltage, V_0 = barrier (junction) potential of the diode, R = series resistance.

I. p-n Junction Diode – Forward Bias

Question 1

A *p-n* junction diode can withstand currents upto a maximum of 10 mA. A resistor $R = 200\ \Omega$ is connected in series with it. The diode, when forward-biased, has a potential drop of 0.5 V across it which is assumed to be independent of current. Find the maximum voltage of the battery used to forward bias the diode.

★ Likely Exam Question



Diode D in series with resistor $R = 200\ \Omega$ and battery V, forward-biased

Given: $I_{\text{max}} = 10 \text{ mA} = 10 \times 10^{-3} \text{ A}, \quad R = 200\ \Omega, \quad V_D = 0.5 \text{ V}$

Formula: $V_{\text{battery}} = I_{\text{max}} R + V_D$

Solution:

By Kirchhoff's voltage law, the battery voltage equals the sum of the drop across the resistor and the (constant) drop across the forward-biased diode:

$$V_{\text{battery}} = I_{\text{max}} R + V_D$$

Substituting the values:

$$V_{\text{battery}} = (10 \times 10^{-3} \text{ A})(200\ \Omega) + 0.5 \text{ V}$$

$$V_{\text{battery}} = 2.0 \text{ V} + 0.5 \text{ V}$$

$$V_{\text{battery}} = 2.5 \text{ V}$$

Final Answer

$V_{\text{battery(max)}} = 2.5 \text{ V}$

Common Mistake: Students often forget to add the diode's forward voltage drop V_D and only calculate $I_{\text{max}} R$, giving an incomplete answer of 2.0 V instead of 2.5 V. Remember: the diode drop is NOT zero even though it is "small" – it must always be included in series-circuit voltage equations.

Question 2

When the voltage drop across a p-n junction diode is increased from 0.65 V to 0.70 V, the change in the diode current is 5 mA. What is the dynamic resistance of the diode?

Given: $\Delta V = 0.70 \text{ V} - 0.65 \text{ V} = 0.05 \text{ V}, \quad \Delta I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$

Formula: $R_{\text{dynamic}} = \dfrac{\Delta V}{\Delta I}$

Solution:

The dynamic (AC) resistance of a diode is the ratio of a small change in voltage to the corresponding small change in current:

$$R_{\text{dynamic}} = \frac{\Delta V}{\Delta I} = \frac{0.05 \text{ V}}{5 \times 10^{-3} \text{ A}}$$

$$R_{\text{dynamic}} = \frac{0.05}{0.005} \ \Omega$$

$$R_{\text{dynamic}} = 10 \ \Omega$$

Final Answer

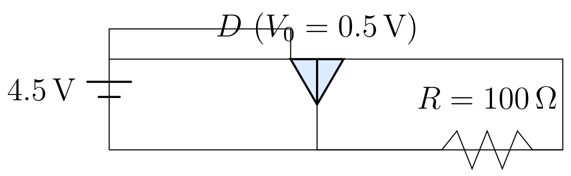
$R_{\text{dynamic}} = 10 \ \Omega$

Common Mistake: Do not confuse dynamic resistance ($\Delta V/\Delta I$, applicable to small changes about an operating point) with static/DC resistance (V/I, the ratio of total voltage to total current). The two are numerically different because the diode's I-V characteristic is non-linear.

Question 3

A p-n junction diode is connected to a series resistor of 100 Ω and a 4.5 V DC battery. The barrier potential at the junction is 0.5 V. Find the current in the circuit.

★ **Likely Exam Question**



Diode D, resistor R = 100 Ω, and 4.5 V battery in series (forward bias)

Given: $V = 4.5 \text{ V}, \quad R = 100 \ \Omega, \quad V_0 = 0.5 \text{ V}$ (barrier potential)

Formula: $I = \dfrac{V - V_0}{R}$

Solution:

The battery voltage is shared between the barrier potential of the forward-biased junction and the voltage drop across the series resistor. Applying Kirchhoff's voltage law:

$$V = IR + V_0 \quad \Rightarrow \quad I = \frac{V - V_0}{R}$$

Substituting the values:

$$I = \frac{4.5\text{ V} - 0.5\text{ V}}{100\ \Omega} = \frac{4.0\text{ V}}{100\ \Omega}$$

$$I = 0.04\text{ A}$$

$$I = 40\text{ mA}$$

Final Answer

$I = 40$ mA

Common Mistake: A very common exam error is to use the full battery voltage (4.5 V) directly in $I = V/R$, ignoring the barrier potential drop across the junction. The barrier potential must always be subtracted first, since it represents the energy needed to overcome the depletion-region potential barrier before current can flow.